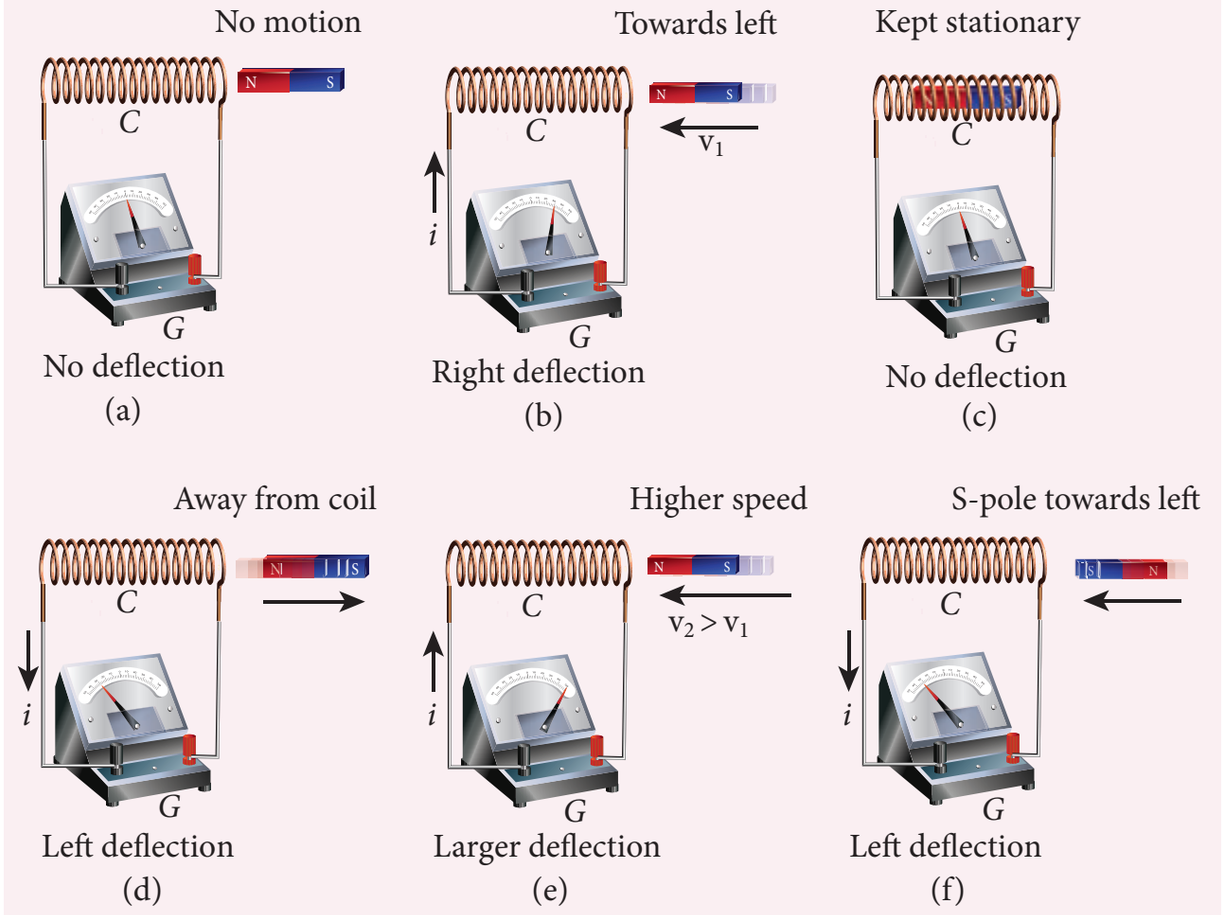


Figure 4.2 Faraday's first experiment

If the magnet is kept stationary and the coil is moved towards or away from the coil, similar results are obtained. It is concluded that whenever there is a relative motion between the coil and the magnet, there is deflection in the galvanometer, indicating the electric current setup in the coil.

Second Experiment

Consider two closed circuits as shown in Figure 4.3(a). The circuit consisting of a coil *P*, a battery *B* and a key *K* is called as primary circuit while the circuit with a coil *S* and a galvanometer *G* is known as secondary circuit. The coils *P* and *S* are kept at rest in close proximity with respect to one another.

If the primary circuit is closed, electric current starts flowing in the primary circuit. At that time, the galvanometer gives a momentary deflection (Figure 4.3(a)).

After that, when the electric current reaches a certain steady value, no deflection is observed in the galvanometer.

Likewise if the primary circuit is broken, the electric current starts decreasing and there is again a sudden deflection but in the opposite direction (Figure 4.3(b)). When the electric current becomes zero, the galvanometer shows no deflection.

From the above observations, it is concluded that whenever the electric current in the primary circuit changes, the galvanometer shows a deflection.

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